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SYNTHESIS LAB



evolution of red blood cell shape in fishes

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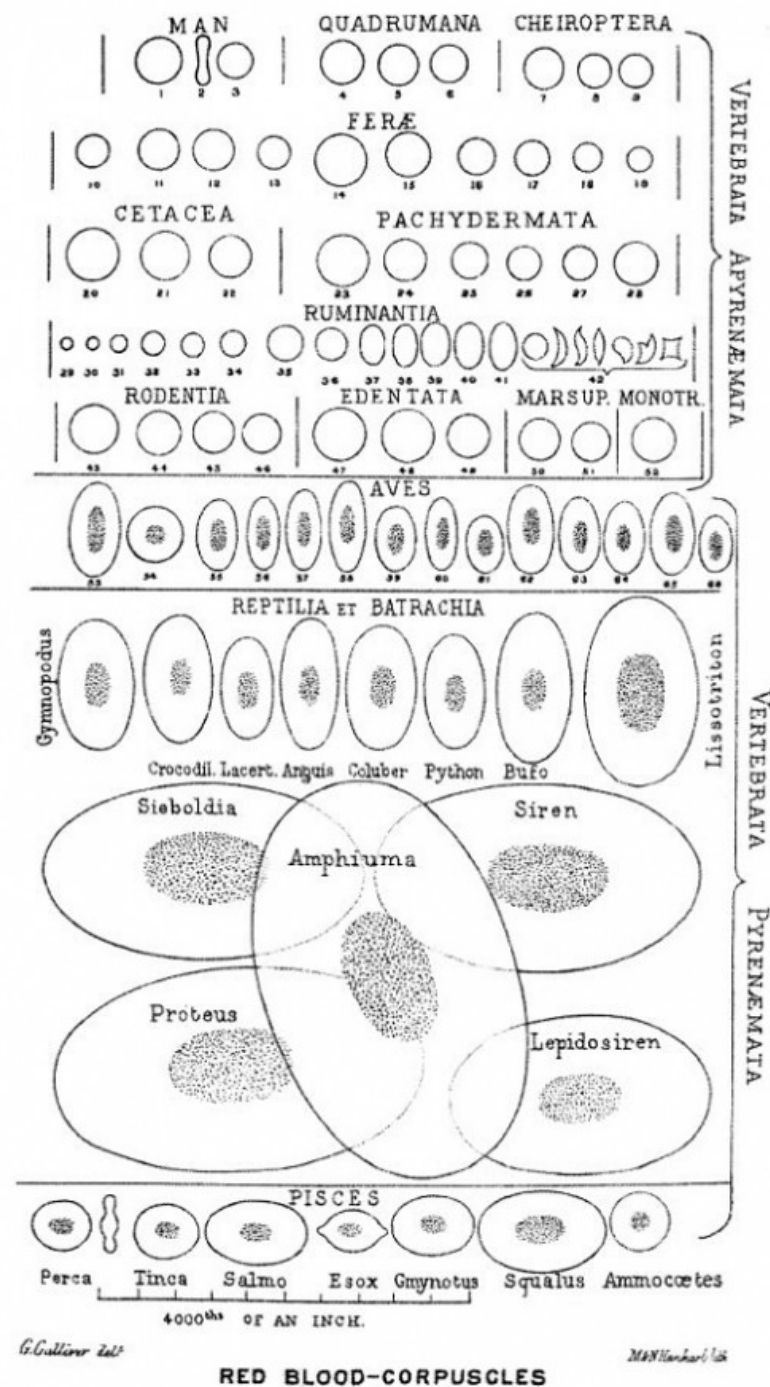
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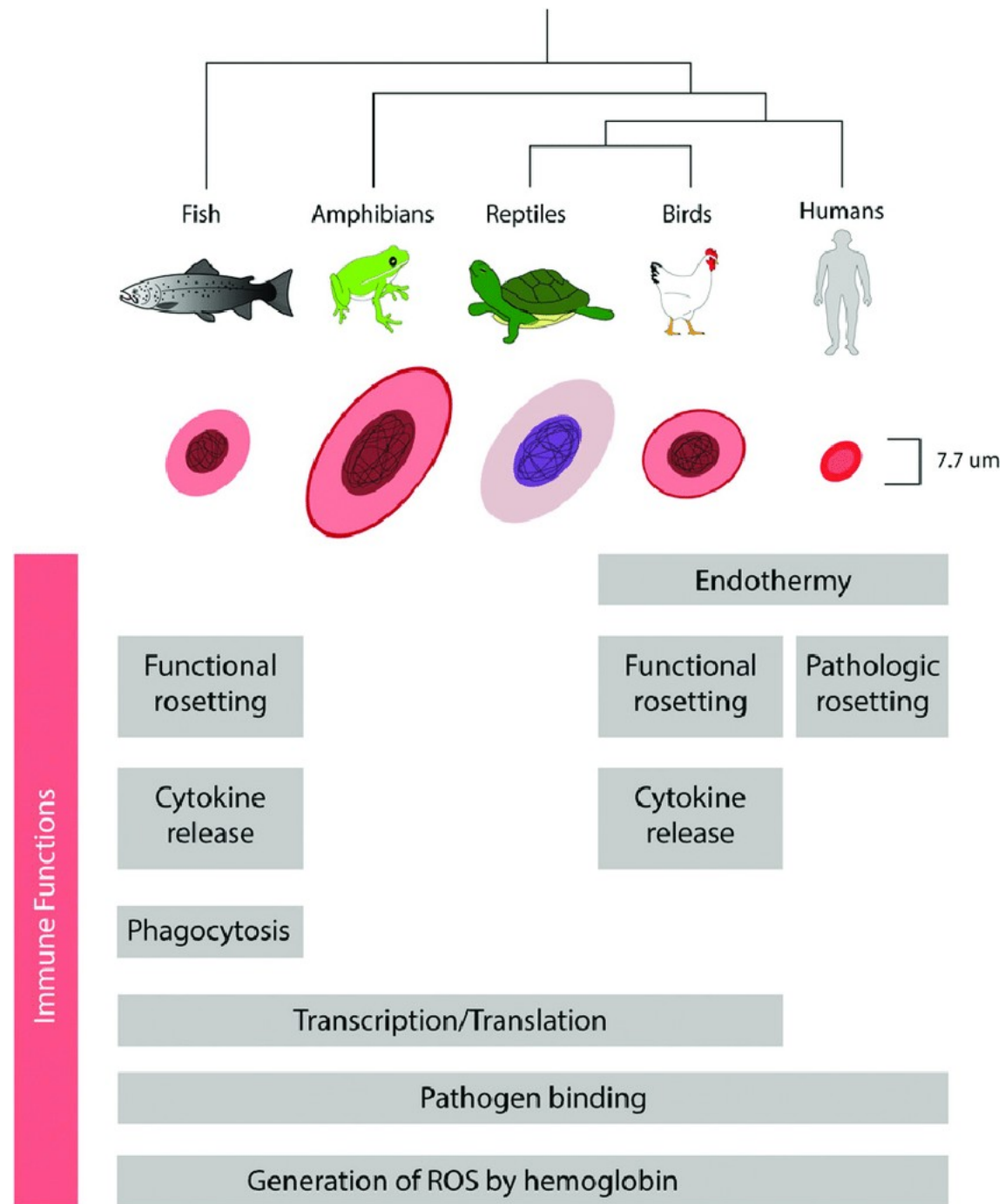
The evolution of red blood cell shape in fishes

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The shape of RBCs varies among vertebrate clades and this has been known for quite some time



Mammals are the only lineage whose RBC lost the nucleus. This has consequences to the immunological functions of RBCs



Introduction

- Several factors can affect the composition and shape of RBC in fish
 - Health status
 - Amount and dynamics of hemoglobin
 - Cell maturation stage
 - Environmental factors
- Larger cells are more deformable than smaller ones
- Temperature and body mass => increase cell volume
- Larger nucleus => genome size / metabolism

What is lacking?

- Little is known about which factors are involved in the variation of RBC volume throughout the fish tree of life and its implications
- The relative role of phylogenetic relationships and abiotic factors in determining RBC shape remain unexplored.

Questions

- How RBC shape evolved in fish
 - Species maximum size and their habitat affect RBC shape
 - Phylomorphospace of RBC shape
 - Phylogenetic signal and rate of evolution

Hypothesis

- Species that live closer to the surface would have smaller RBC due to higher surface area to volume ratio
- Smaller species would have larger RBC, because they would have higher metabolic rate
- Closely-related species would be grouped together in the phylomorphospace

Data from the data base

Back to [Animal Genome Size Database](#).
Back to [Cell Size Database](#).

Fish erythrocyte sizes

NOTE: These are size measurements for fish erythrocytes (red blood cells). Erythrocytes in fishes are elliptical, and as such two different diameters are provided: CLD ("cell long diameter") and CSD ("cell short diameter"). NLD ("nucleus long diameter") and NSD ("nucleus short diameter") are also provided where available. In both cases, the diameters were measured from dry smears. These were used to calculate dry cell area (CA) and dry nucleus area (NA) using the equation for elliptical area: $CA = \pi * (CLD/2) * (CSD/2)$. In some cases, area is measured directly using image analysis with a calibration slide, so diameters are not available. MCV ("mean corpuscular volume") is wet volume and is usually measured with a Coulter counter. These different measurements (e.g., CA and MCV) should *not* be compared to each other, so pick only one for any analyses you wish to do, or run them separately. Diameters are in μm , areas are in μm^2 , and volumes are in μm^3 .

Species and taxonomic information	CLD	CSD	CA	Ref	NLD	NSD	NA	Ref	MCV	Ref
SUPERCLASS AGNATHA										
CLASS CEPHALASPIDOMORPHI										
ORDER Petromyzontiformes										
Family Petromyzontidae										
<i>Geotria australis</i>	12.8	12.8	128.68	11					370	11
<i>Lampetra fluviatilis</i>	11.6	11.6	105.68	11					189	11
" "	11.9	11.9	111.27	1						
<i>Lampetra planeri</i>	11.9	11.9	111.27	1						
<i>Petromyzon marinus</i>	11.9	11.9	111.27	1	4.0	4.0	12.37	1	534	13
" "	14.3	14.3	160.61	6					710	6
CLASS MYXINI										
ORDER Myxiniformes										
Family Myxinidae										
<i>Myxine glutinosa</i>	26.2	18.0	370.39	3					1530	3
<i>Myxus petardi</i>	10.2	7.0	56.08	2	3.5	2.5	6.87	2		
SUPERCLASS GNATHOSTOMATA										
CLASS CHONDRICHTHYES										
SUBCLASS ELASMOBRANCHII										
ORDER Carcharhiniformes										
Family Carcharhinidae										
<i>Carcharhinus brachyurus</i>			175.36	14			31.92	14		

38 species from 21 orders of Teleostei

12 species of four orders of Elasmobranchii

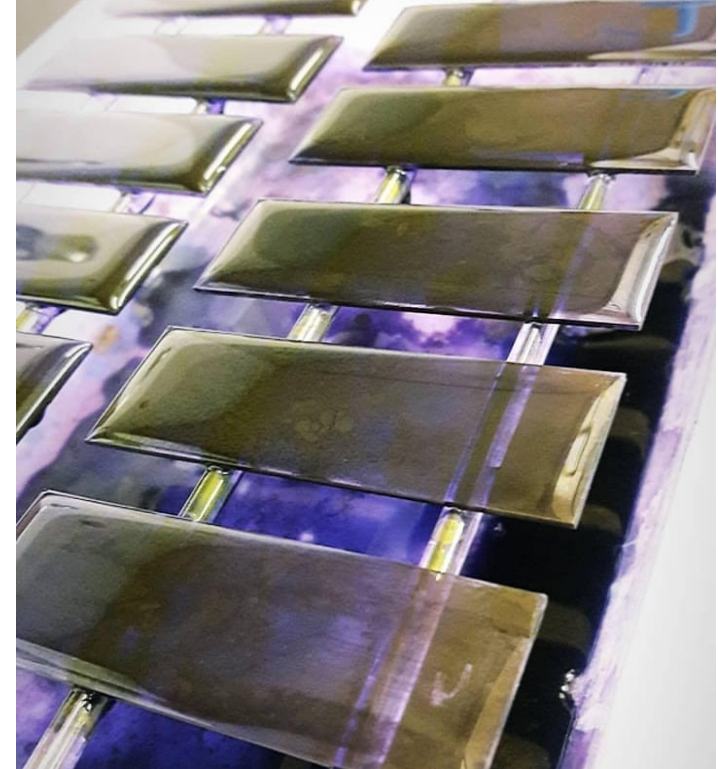
Four variables:

cell largest and smallest diameter
and nucleus largest and smallest
diameter

Methods

Data from field-collected fish

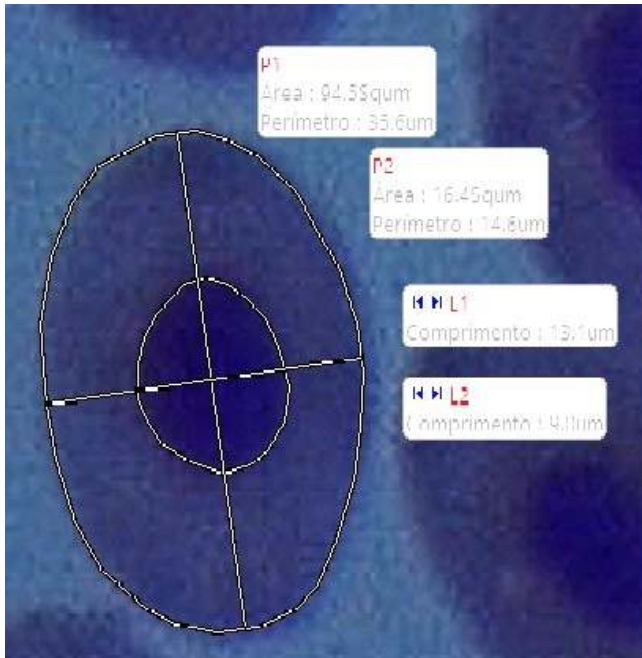
- Blood smears for 15 species of Teleosts
- May-Grunwald-Giemsa-Wright (MGGW)
- 5 individuals per species
- 50 cells per animal => 250 cells per species



After Rosenfeld (1947) and
Tavares-Dias e Moraes (2003)

First-order stereological variables

largest and smallest diameter (μm)
of cells and nucleus



Second-order stereological variables

- Cell volume (μm³)

$$\frac{4}{3\pi}r^3$$

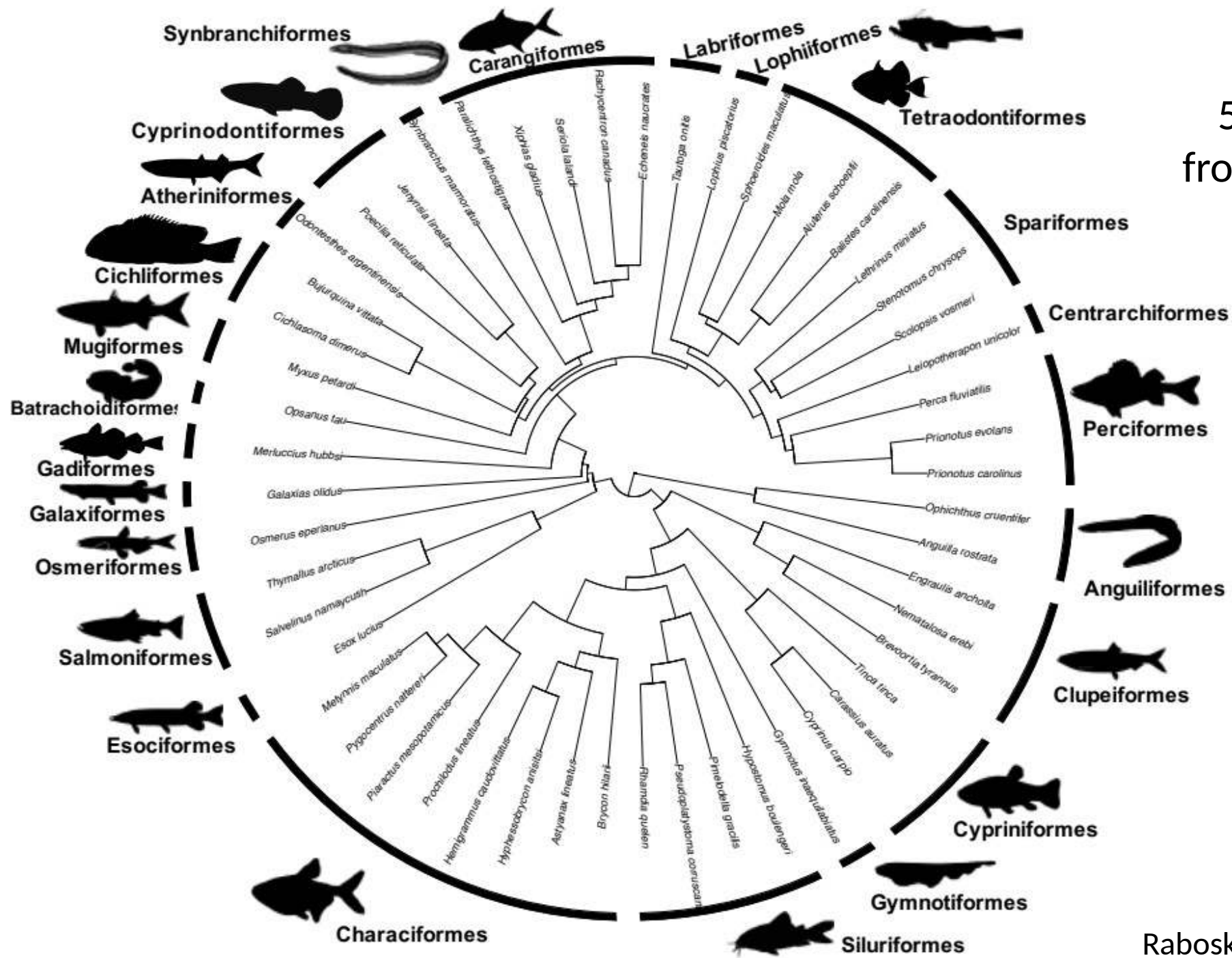
- Cell and nucleus area (μm²);

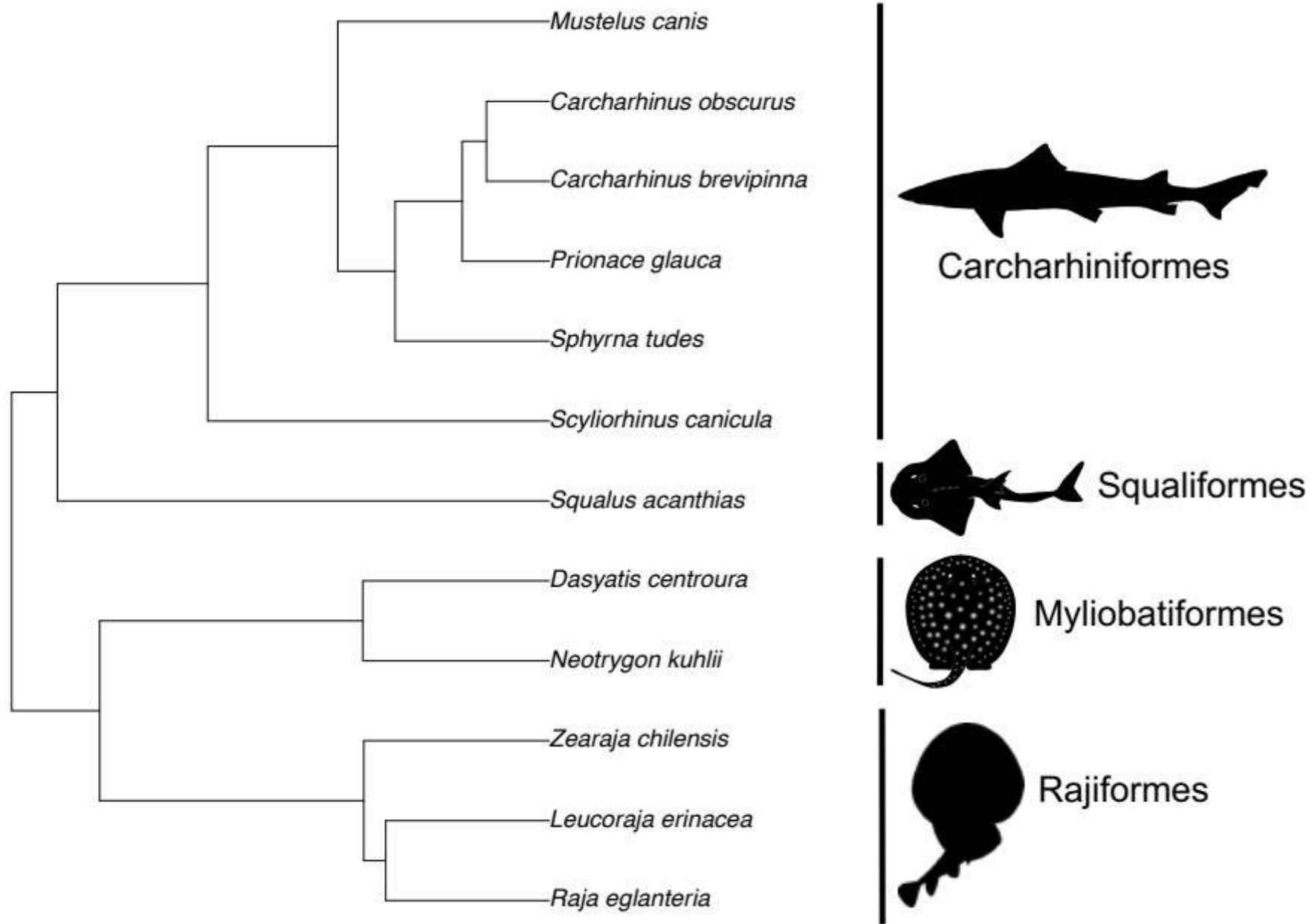
- Nucleus:cytoplasm ratio (%)

- Nucleus volume (μm³)

$$\frac{CV * N:C}{100}$$

- Shape factor $\frac{4\pi A}{P^2}$

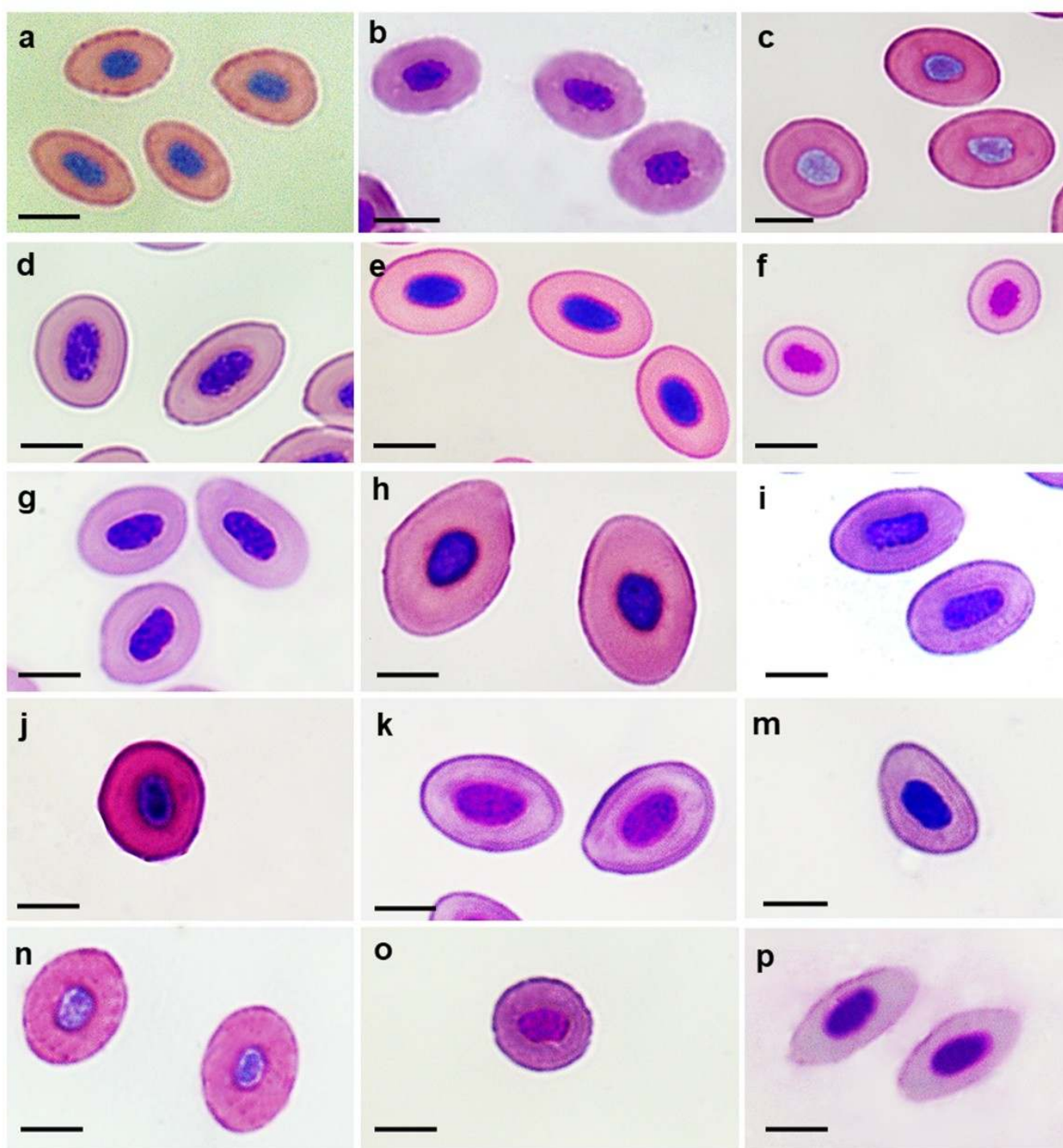




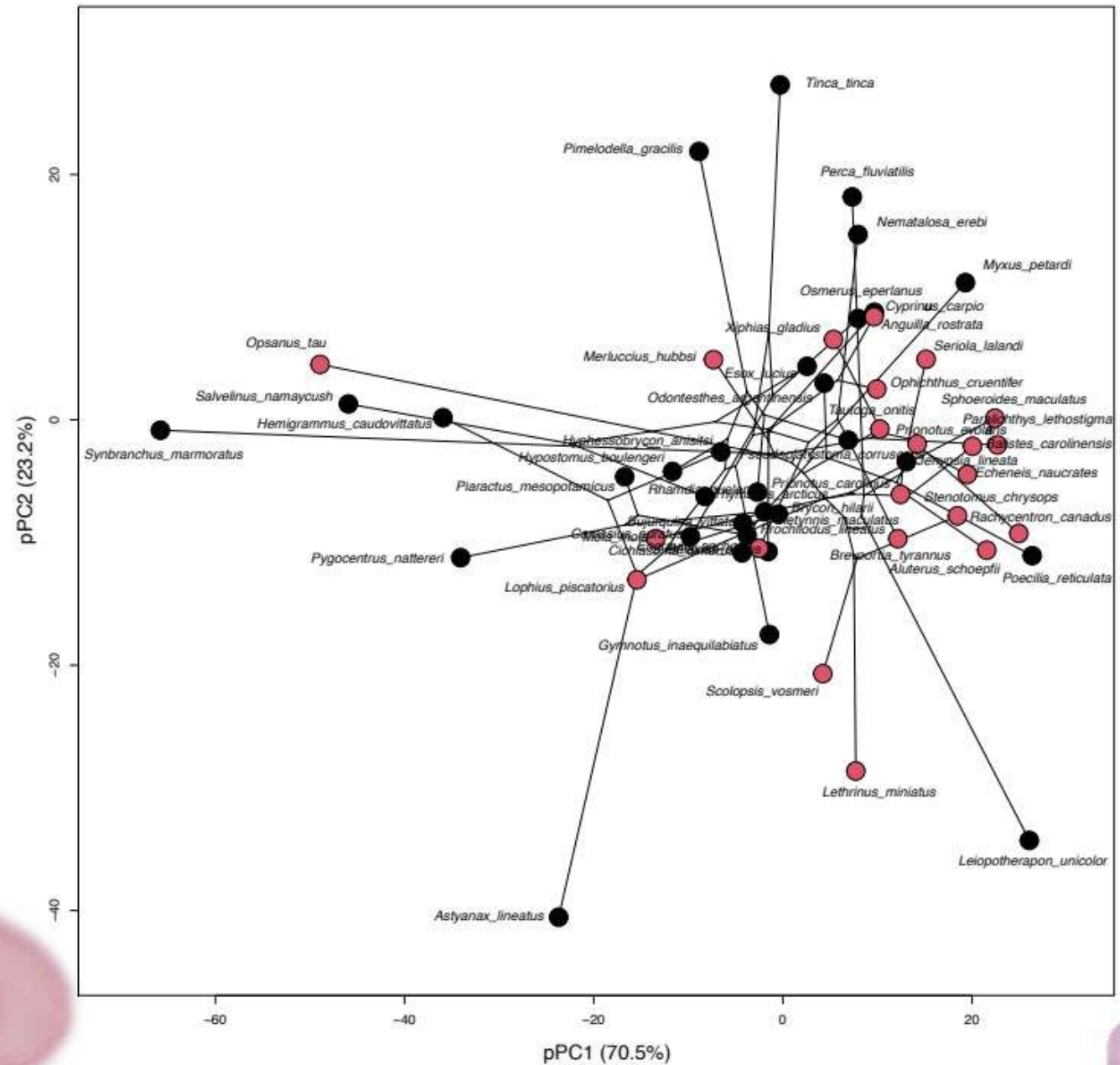
Methods

- Phylogenetic Comparative methods for multivariate data
- Phylomorphospace: Phylogenetic Principal Component Analysis (pPCA)
- Phylogenetic signal: K_{mult} with all PCs
- Multivariate PGLS for testing the relationship between RBC shape ~ habitat + maximum size
- Rate shift: RRphylo for estimating evolutionary rate in tips and nodes, and testing for rate shifts throughout the phylogeny

RESULTS

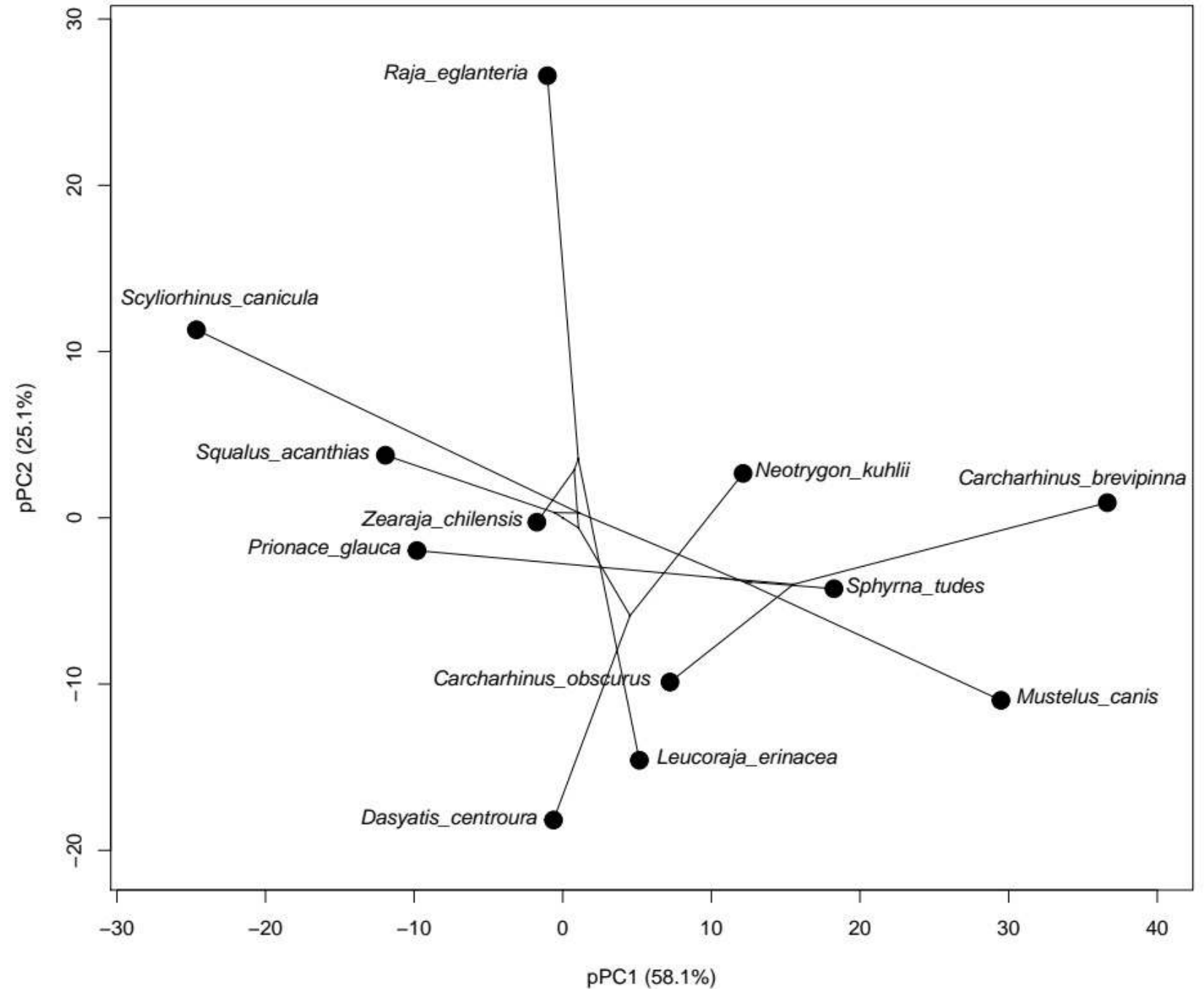


1000x)

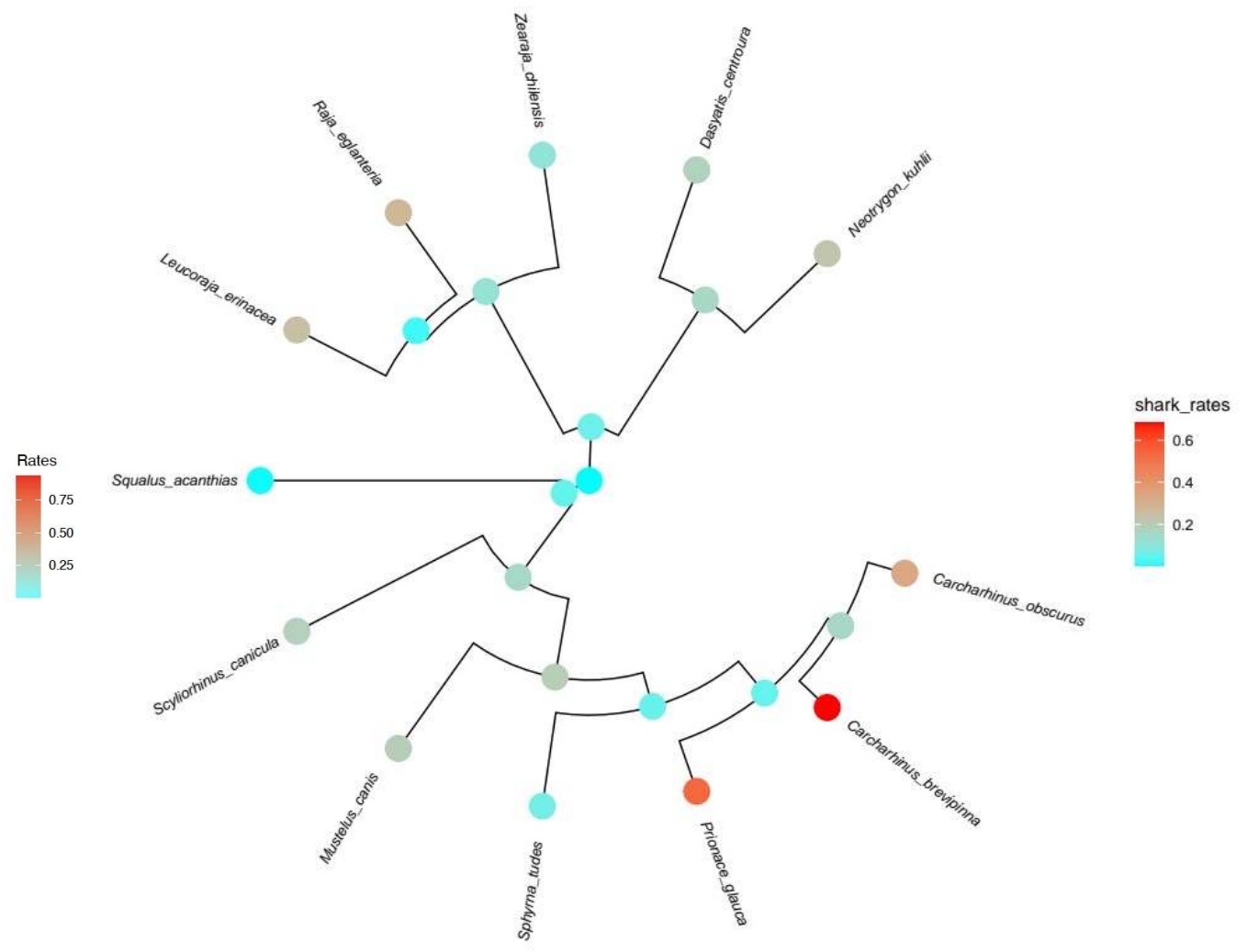
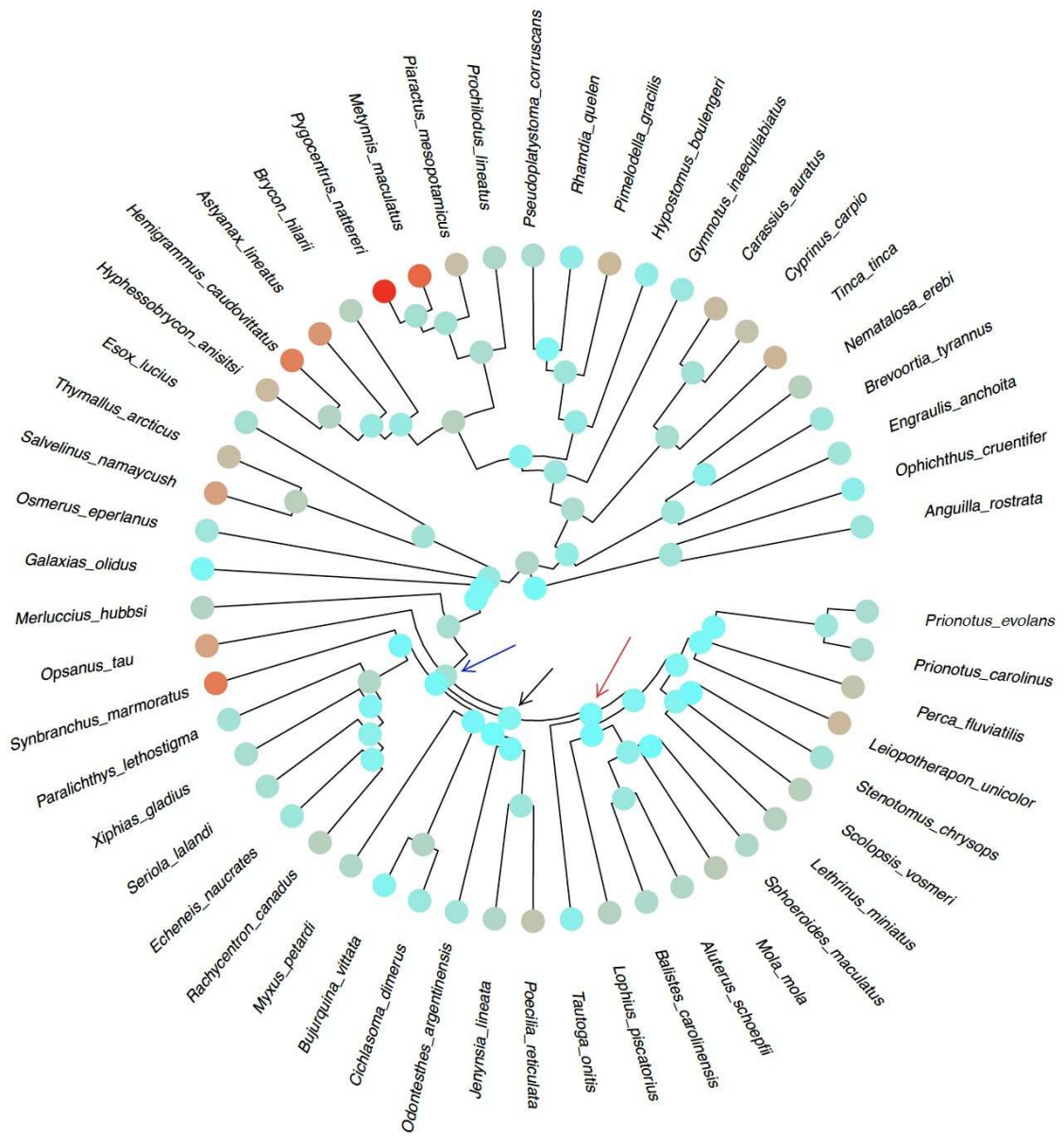


Results

$K_{\text{mult}} = 0.4862$;
Effect size = 0.623;
 $P = 0.4862$



Results: RRphylo



Results = PGLS

Table 1. Results of the Phylogenetic Generalized Least Squares for the effect of maximum body size and habitat on the shape of red blood cells. Legend: DF=Degrees of Freedom, SS=Sum of Squares, MS=Mean Squares, Z= z-score.

Teleosts							
	DF	SS	MS	R ²	F	Z	P
Habitat	6	72.874	12.1456	0.23357	2.2955	1.3278	0.095
Maximum size	1	1.025	1.0249	0.00328	0.1937	-1.0104	0.845
Residuals	45	238.101	5.2911	0.76315			
Total	52	312.000					
Sharks							
	DF	SS	MS	R ²	F	Z	P
Habitat	3	22.608	7.5359	0.34254	1.4053	0.62147	0.288
Maximum size	1	5.854	5.8538	0.08869	1.0916	0.40632	0.378
Residuals	7	37.538	5.3626	0.56876			
Total	11	66.000					

Conclusion

- RBC of fish vary greatly in shape, but neither habitat nor body explain it.
- RBC shape may not be entirely determined by partial pressure of dissolved oxygen in water
 - RBC in fish have their own organelles and can rapidly respond to changes in O_2 levels
- Future studies should use quantitative genetic approaches to explore the underlying genetic basis for RBC morphology, possibly using geometric morphometrics and experiments

Thank you !

Feel free to contact me if you have
any further questions

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